

- 19** A piece of wire of original length L , has a resistance of R . It is then melted and made into a new wire of length $1.7 L$.

What is the resistance of the new wire?

A $0.59 R$

B R

C $1.7 R$

D $2.9 R$

Ans: D

Since volume remains constant,

$$\begin{aligned}A_{old}L_{old} &= A_{new}L_{new} \\A_{new} &= \frac{L}{1.7L}A_{old} = 0.59A_{old} \\R_{new} &= \rho \frac{L_{new}}{A_{new}} = \rho \frac{1.7L}{0.59A_{old}} = \frac{1.7}{0.59}R = 2.9R\end{aligned}$$

20. In the circuit below, 3 identical resistors of resistance $1.0 \text{ k}\Omega$ are connected to a cell of 1.2 V